



American International University- Bangladesh (AIUB)

Faculty of Engineering

Course Name: Microprocessor and Embedded System

Course Code: EEE 4103

Semester: Fall 2023-24

Term: Mid

Faculty Name: Tahseen Asma Meem

Assignment #: 02/02

Course Outcome Mapping with Questions

Item	COs	POIs	K	P	A	Marks	Obtained Marks
Q1-4	CO1	P.a.4.C3	K4	P1, P3, P7		5×4	
Total:						20	

CO/ CLO Number	CO/CLO Statement	K	P	A	Assessed Program Outcome Indicator	BNQF Indicator	Teaching- Learning Strategy	Assessment Strategy
CO1	Apply knowledge of microprocessors and microcontrollers to configure different modules of a microprocessor or microcontroller-based system as per given specifications and perform in-depth analysis to optimize the performance of the designed system utilizing the basic concepts and properties of a microcontroller.	K4	P1, P3, P7		P.a.4.C.3	FS.2	Lecture/ Discussion/ Q&A	Midterm Assignment 2

Student Information:

Submission Date:	02/11/2023				
Student Name:	RIAD AL HASAN				
Student ID #:	22-46732-1	Department: CSE		Section: K	

Marking Rubrics (to be filled by Faculty):

Problem #	Excellent [5]	Proficient [4]	Good [3]	Acceptable [2]	Unacceptable [1]	No Response [0]	Secured Marks
	Detailed unique response explaining the concept properly and answer is correct with all works clearly shown.	Response with no apparent errors and the answer is correct, but explanation is not adequate/unique.	Response shows understanding of the problem, but the final answer may not be correct	Partial problem is solved; response indicates part of the problem was not understood clearly or not solved.	Unable to clarify the understanding of the problem and method of the problem solving was not correct	No Response/ copied from others/identical submissions with gross errors/image file printed	
1							
2							
3							
4							
Comments						Total marks (20)	

1. In an ultimate fun factory, you as a player want to play an interesting game named as Laser Tag. In the preparation room, every player was requested to wear a vest to get identified in the dark maze play zone. The vest contains 1 light in the middle of the vest in front, 2 lights on 2 shoulders (left and right) and 1 light on the behind of the vest and, these 4 lights flash together periodically. It was found that the system is built with an Arduino Uno at the heart. Interestingly, the Arduino was already set up to consume as little power as possible not to strain the vest's efficiency. Now, prepare a program in the Arduino Uno platform to control the 4 lights mounted on the vest so that all the lights flash together every 15s to indicate every player. [5]
2. Prepare a flowchart to explain the flow of logic for the program in Q1 so that the vest can be modified easily if the vest requires any maintenance after every game is over. [5]
3. Prepare an algorithm to explain the flow of logic for the program in Q1. [5]
4. On new year's eve, the Hatirjheel authority is planning to shoot high powered high frequency laser light waves as pulses into space for research purposes. These pulses are to be shot at a **30 μ s** interval. The authority has an **ATMega328P MCU** available and decided to use it to count the time. Every **30 μ s** a timer interrupt will be triggered by the MCU and the pulse will be shot. For this purpose, considering the power consumption and necessity, it has been decided that the Timer0 interrupt will be used. If the MCU is running at 16 MHz, prepare a program that triggers Timer0 interrupt every **30 μ s**. Timer0 is an 8-bit timer, and the available pre-scalers are 1, 8, 64, 25, and 1024.
[Hint: **Bits 0 to 2** of the **TCCR0B** register must be set. These bits correspond to the pre-scaler to be used. The pre-scaler selection bits are shown in the **table below**. Output Compare Register 0A: **OCR0A** holds the necessary Timer0 count to achieve the required delay. **Bit 1** from **TIMSK0** is **OCIE0A** – this bit must be set to enable comparison match A. Any bits not mentioned here are to be considered as 0. All registers mentioned are 8-bit registers.] [5]

CSx2	CSx1	CSx0	Prescaler
0	0	1	1
0	1	0	8
0	1	1	64
1	0	0	256
1	0	1	1024

This Individual Assignment should be handwritten and then converted to pdf, the file should be saved in the format ID.pdf and then submitted to the following Submission link: <https://forms.office.com/r/0PyGHng9q6> the submission link will close on 3rd November 2023.

Answer-1

①

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Program

```
int lightpinfront = 2;  
int lightpinleft = 3;  
int lightpinright = 4;  
int lightpinback = 5;  
unsigned long previousMillis = 0;  
const long interval = 15000;
```

```
void setup() {  
    pinMode(lightpinfront, OUTPUT);  
    pinMode(lightpinleft, OUTPUT);  
    pinMode(lightpinright, OUTPUT);  
    pinMode(lightpinback, OUTPUT);  
  
    digitalWrite(lightpinfront, LOW);  
    digitalWrite(lightpinleft, LOW);  
    digitalWrite(lightpinright, LOW);  
    digitalWrite(lightpinback, LOW);  
}
```

```
void loop() {  
    unsigned long currentMillis = millis();  
    if (currentMillis - previousMillis >= interval)  
    {  
        // ...  
    }  
}
```


previousMillis = currentMillis ;

digitalWrite(lightpinFront, High);

digitalWrite(lightpinLeft, High);

digitalWrite(lightpinRight, High);

digitalWrite(lightpinBack, High);

delay(500);

digitalWrite(lightpinFront, Low);

digitalWrite(lightpinLeft, Low);

digitalWrite(lightpinRight, Low);

digitalWrite(lightpinBack, Low);

}

}

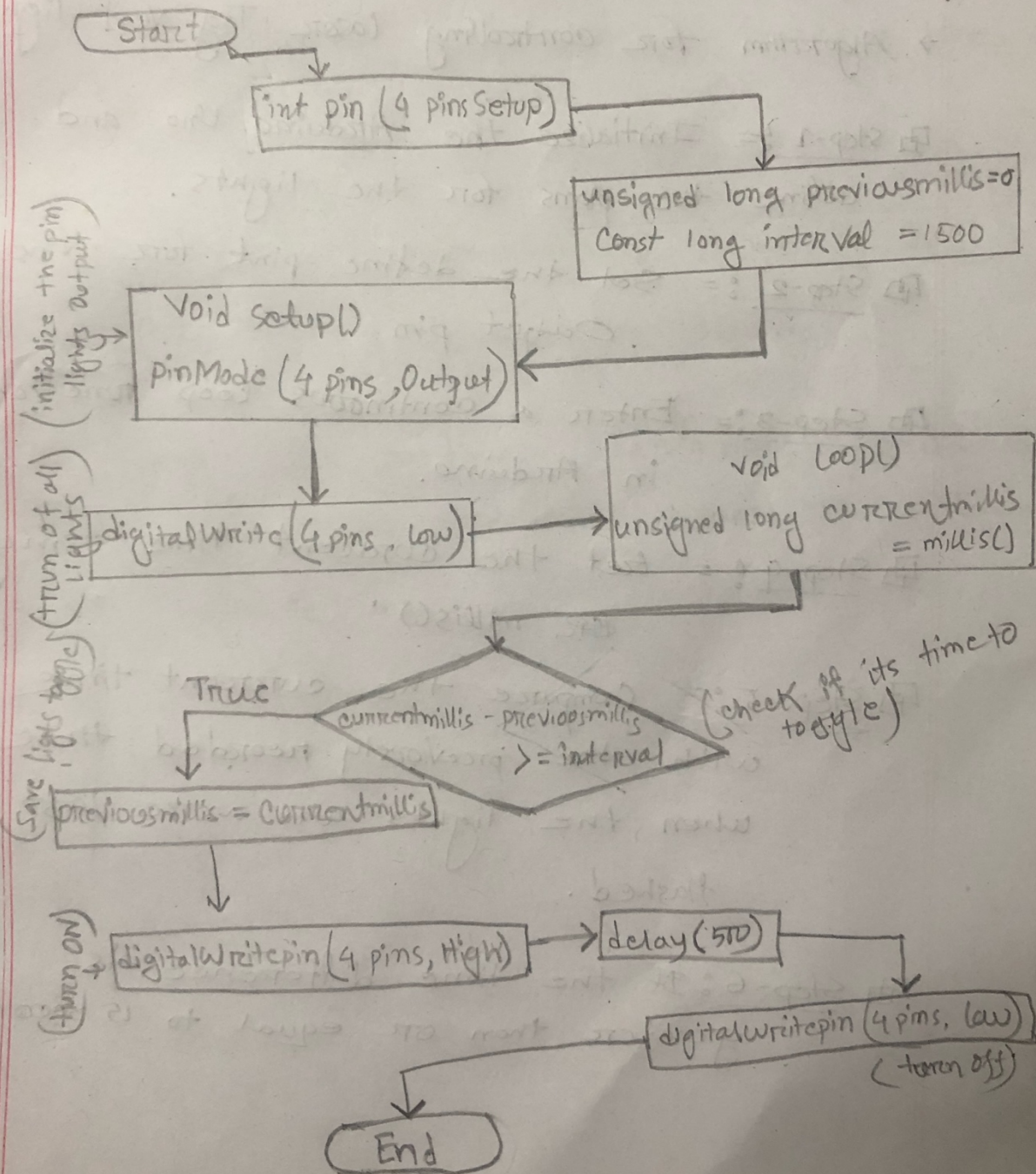
When the Arduino is powered on, it will flash all four lights together every 15 seconds, indicating the player into the laser tag game.

③

Answer-2

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Flowchart for Questions No-1



Answer-3

Algorithm for Question-1

→ Algorithm for controlling Laser Tag Vest light

▣ Step-1 := Initialize the Arduino Uno and define the pins for the lights.

▣ Step-2 := Set the define pins for the Output pin

▣ Step-3 := Enter a continuous loop function in Arduino.

▣ Step-4 := Get the current time using ~~the~~ millis().

▣ Step-5 := Compare the current time with the previously recorded time when the lights were last flashed.

▣ Step-6 := If the time difference is greater than or equal to 15 seconds

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- Record the time as the new previous time.
- Turn on all four lights.
- Delay for a short period to keep the lights on.
- ~~the~~ Turn off all four light.

Step -7 : continue looping, repeating the light flashing every 15 seconds.

The loops will continue indefinitely, periodically flashing the lights together every 1500 milliseconds to indicate the prayer.

⑥

Answer-4

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Calculation (30 K μ s)delay = 30 μ s ;

prescaler = 64 ;

System-frequency = 16 MHz ;

$$\text{frequency} = \frac{16 \text{ MHz}}{64} = 250 \text{ KHz ;}$$

$$\text{Time-period, } T = \frac{1}{\text{frequency}} = \frac{1}{250 \text{ KHz}} = 4 \mu\text{s ;}$$

$$\text{Timer count} = \frac{\text{Required Delay}}{\text{Time period}} - 1$$

$$= \frac{30 \mu\text{s}}{4 \mu\text{s}} - 1$$

$$= 6.5 \approx 7$$

bool LED-STATE = true ;

void setup() {

pinMode (13, Output) ;

cli() ;

TCCR0A = 0 ;

TCCR0B = 0 ;

TCCR0B = B00000010 ;

TIMSK0 = B000000010 ;

OCR0A = 7 ;

} Serial ;


```
void loop() {
```

```
  ISR (TIMER0_COMPA_vect)
```

```
  {
```

```
    TCNT0 = 0;
```

```
    LED_STATE = !LED_STATE;
```

```
    digitalWrite(13, LED_STATE)
```

```
  }
```

```
}
```